

Indoor Plant Grower

Science and Home

Science and Agriculture

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Introduction

With an ever-increasing population and decrease in agricultural land, urban farming and growing your produce are becoming very important. However, growing plants at home without technical knowledge is a very tough task. Growing plants at home means fulfilling the constant needs of the plants, i.e., regular watering and providing ample sunlight according to the plant growth. Moreover, there is also the problem of overwatering, underwatering, pests, and harmful external environment. Many people fail to check on their plants constantly. A single day of negligence will cause their plant to die. How can a busy person grow his plants properly then?

The Indoor Plant Grower helps with these situations. The Indoor Plant Grower is an IoT-based electronic solution for growing plants. With this system, you only need to plant a seed. After that, the system itself does all the work. The system monitors atmospheric conditions and plant growth. It monitors soil humidity and waters the plant accordingly. It also has a growth light which creates an artificial sun without the harmful UV rays. Finally, you can monitor your plant growth from your phone. Thus, your plant can grow indoors without help.

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Working

An Arduino Uno is used to control the Indoor Plant Grower. The Arduino Uno is attached to the following :

- Soil Humidity Sensor YL-69: This sensor detects soil humidity level
- DHT11: This sensor detects the atmospheric temperature and humidity levels
- Adafruit NeoPixel Strip: This LED strip creates the grow light
- Ultrasonic Sensor HC-SR04: This sensor measures the distance between the plant and the top, thus monitoring plant growth
- ESP32 Microcontroller: This microcontroller acts as a wifi shield for the Arduino Uno. This microcontroller connects and writes to the website.
- 5v DC Water Pump: This pump waters the plant whenever required
- Relay Module: The relay allows the additional power supply to the water pump
- 0.96in OLED Display: This display shows plant information directly on the Plant Grower

The three sensors, together, monitor all the essentials for a plant, i.e., Soil Humidity, Ambient Atmosphere, and Plant Growth. The sensors provide a well-rounded range of inputs that determine overall plant growth.

A Relay is attached to the Water Pump allowing the pump to use external power while being controlled by the Arduino. Providing water is the main requirement for growing plants. However, the quantity of water to be provided varies from time to time based on

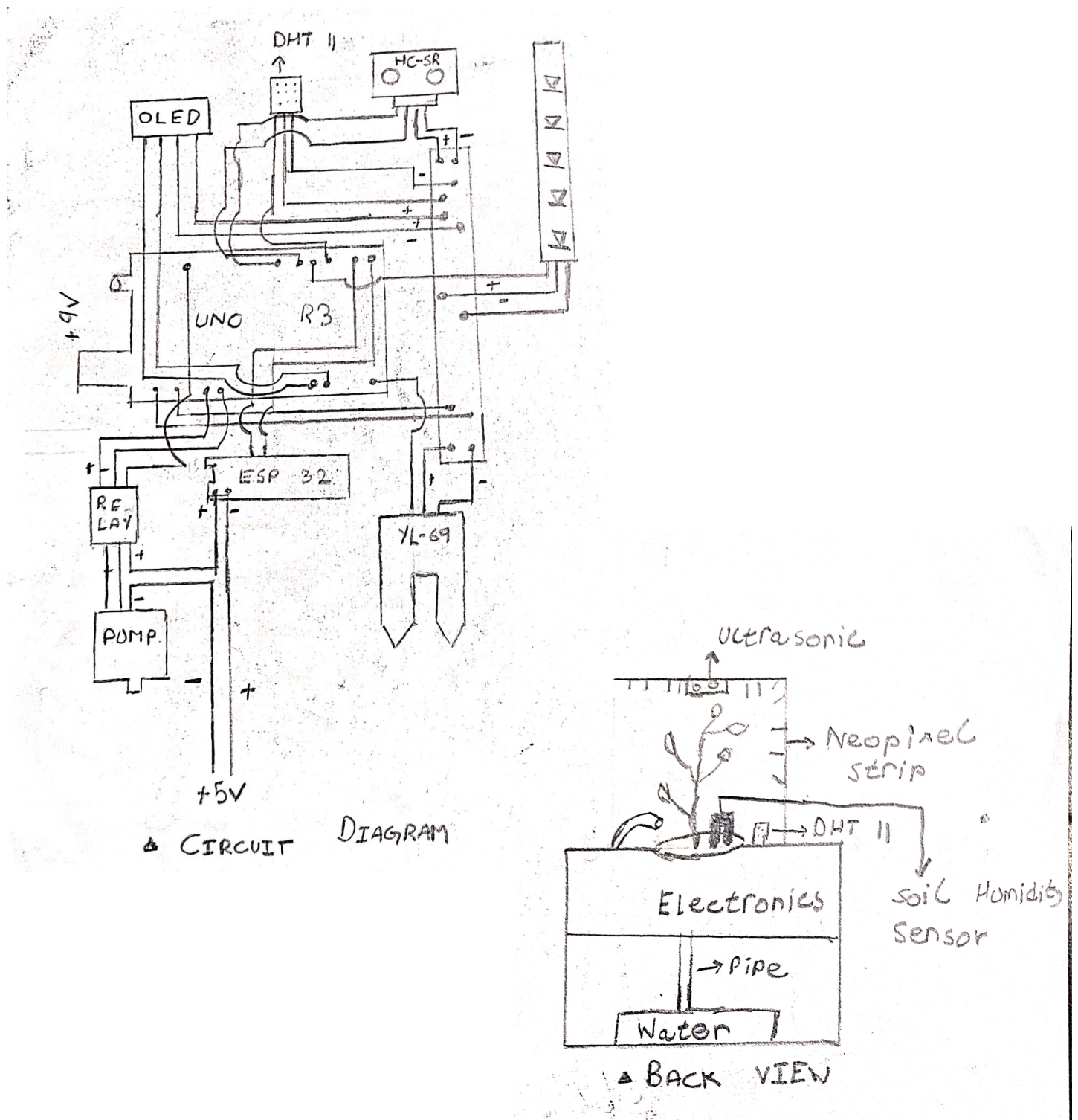
plant requirements. The pump is programmed to fulfil only the current water requirements, thus preventing drying and waterlogging in the plant.

The NeoPixel is used to create an artificial sun, which is the final step for making the plant grow entirely indoors. The NeoPixel provides colours on the red and blue spectrums with a shade of green which is the ideal light for photosynthesis. The NeoPixel also mimics the sun to meet the plant's brightness requirements.

The ESP32 and Relay are connected to a separate 5v DC power supply, while the Arduino uses a 9v DC power supply from the socket. The Arduino transfers the sensor data to the ESP32. The ESP32 then puts the data onto a website where it is displayed in a graphical manner making the plant accessible anytime, anywhere.

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Diagram



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Feasibility

The Plant Grower is constructed using waterproof PVC boards. This frame makes the system sturdy, lightweight, and easy to transport. This frame also keeps all the electronics safe from water and dust. The project is compact, allowing the growth of plants in small apartments. The pot and water reservoir are not attached to the frame and thus can be adjusted to meet the plant and consumer demands.

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Bibliography

Project Ideas :

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- Herb Box Ecosystem : <https://www.hackster.io/walwode/herb-box-eco-system-7c51b3>
- Arduino Project Hub : <https://create.arduino.cc/projecthub>
- Plant Growth Light : <https://www.youtube.com/watch?v=pQGSjXb6EBo>

Tutorials and Instructions :

- Arduino Docs : <https://docs.arduino.cc/>
- ESP32 Docs : <https://docs.espressif.com/projects/esp-idf/en/latest/esp32/>
- Random Nerd Tutorials : <https://randomnerdtutorials.com/>
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